

MVP scope — proposal

Status: proposed, not agreed. Closing question S1 turns this into the build plan. Until then, treat every line as a suggestion.

The MVP test

The MVP is the smallest set of features that closes the loop end to end for one asset class on one site:

```
a fault is detected → a case opens → it is explained with evidence
→ the checks narrow it to a root cause → a work order is created
→ the work is done → the platform proves it worked
```

The case is the step the first draft of this document missed. A fault does not become a work order in one hop: somebody has to answer questions, and some of those answers come back "could not check". That middle is where the work actually happens, and it is RC1 – RC19 .

Anything that does not sit on that path is deferred, however attractive it is. A demo that answers questions beautifully but cannot prove a repair worked has not demonstrated the product.

What is in — 94 of 147 features

Domain	In MVP	Why
Synex Copilot	C1–C12	The whole point. Ask, resolve context, route, gather evidence, explain, recommend, refuse honestly, draft, route approval, hold task state, respect scope, cite sources
Conversation shell	C15–C20	The Copilot is the product, so the shell is not optional: thread memory on every route, a conversational front door, read-only data lookup, route transparency, export, and the skill registry all of it routes through
Honesty discipline	C21–C26	What stops a fluent answer from being a confident wrong one: every number a value or a stated absence, every artefact carrying its data window, untrusted periods marked and honoured, figures drawn only from the evidence pack, four response modes graded by how much the evidence can actually settle, and per-signal provenance so a fabricated measurement can never read as a real one
Case Resolution	RC1–RC19	The lifecycle between a named fault and a closed work order: state machine, curated checklists, capability routing, findings with an explicit cannot-check, the blocking gate, root cause with corrective and preventive actions, three escalation routes, seeding that is both safe to re-run and actually run, ageing, measured-versus-estimated, stored readings offered only as a confirmation, correlation before escalation so one problem does not become five work orders, differential narrowing with an elimination audit, honest exhaustion, the work order escalation produces, and a deterministic assignee
FDD	F1–F8, F10–F11, F14–F17	The six models, residuals, gates, isolation path, sensor-bias detection, honest ambiguity, NO_DIAGNOSIS, model health, quarantine, ΔT check, per-asset reference bands, cross-signal physical plausibility, and one severity scale
Work Orders	W1–W4, W8–W10	Create from chat or fault, carry the evidence, deterministic priority, capture findings, cannot close unproven, reopen on failure
Verification	V1–V4, V6	Post-work residuals under valid gates, persistence, fault clear, PASS/FAIL/UNKNOWN
Reports	R1, R3, R5, R10	Ask, explain a change, drill to source, reconcile against source
Knowledge	K1, K5	SOP retrieval with the source shown
Asset	A1	One equipment story
Roles	U3, U6–U8	Four personas, because the loop needs a fault judged, worked, approved and governed
Energy	E1	The efficiency baseline the FDD proxy is measured against
Evaluation	EV1–EV4	A demonstrator that cannot be shown to be honest is worth little: a golden case set, a gate that fails the build on dishonest answers, dimensions no overall score can trade away, and tests of the gate itself
Safety	S1, S4, S6	Block safety-critical actions; safety answers only from the SOP; and a response class for faults answered by stopping the machine rather than by raising a work order

Domain	In MVP	Why
Control Plane	G1–G6	Identity and scope per turn, risk class, approvals, tool gateway, idempotency, audit

The shape of the cut, and why it stopped growing

The table below is generated from the register, so it cannot drift from what is actually being built. Read the *shape* of it rather than the totals.

Domain	In	Out	In the cut
Synex Copilot	24	2	C1–C12, C15–C26
Case Resolution	19	0	RC1–RC19
Reliability & FDD	14	3	F1–F8, F10–F11, F14–F17
Work Orders	7	5	W1–W4, W8–W10
Control Plane	6	2	G1–G6
Verification	5	2	V1–V4, V6
Reports	4	6	R1, R3, R5, R10
Evaluation	4	0	EV1–EV4
Roles	4	4	U3, U6–U8
Safety	3	3	S1, S4, S6
Knowledge	2	4	K1, K5
Asset Intelligence	1	4	A1
Energy & Cost	1	3	E1
Planning	0	5	— deferred whole
Inventory	0	4	— deferred whole
Alerts	0	6	— deferred whole
Total	94	53	of 147 registered

89 of the 94 are P0. Almost nothing in the cut is optional; the 5 that are not are there because the loop reads badly without them, not because they are nice to have. Deferred work splits 45 to Phase 2 and 8 to Phase 3.

2 domains are in whole — Case Resolution, Evaluation — and **3 are out whole** — Planning, Inventory, Alerts. That is the shape of the decision, not an accident of counting. A demonstrator has to close the loop *completely* and does not have to be *broad*, so the middle of the loop is taken entire while domains that are genuinely valuable, but not on it, are taken not at all.

The cut has grown four times, and each time for the same reason

Cut	When	What was found
51 of 101	the first proposal	—
69 of 122	after reviewing the Thermynx flows	The loop went from a named fault straight to a work order. That hop does not exist: somebody answers questions first, and some answers come back "could not check" (D-002)
79 of 132	after the FDD discovery pass	Four gaps that had each already produced a real incident there — C23 , RC9 , RC10 , S6 (D-004)
90 of 143	after the sequencing brainstorm	22 detected episodes, including the only two critical , never reached the queue — RC17 (D-007)
93 of 146	after reading our own data	Evidence sitting unread in the row the model had just read, two severity scales disagreeing, and a signal the simulation invented — RC18 , F17 , C26 (D-008, D-009)

Every increase came from finding a defect in a system that had already been built and run, never from imagining a feature. That is the useful pattern, and it is also the reason to believe the growth has stopped: the last four additions were **rules** — each a paragraph of behaviour attached to a component already in the cut — rather than new subsystems. Running out of subsystems and finding only rules is what the end of a gap hunt looks like.

This is what question S1 asks. Not "is 93 the right number", but "are we done adding?" Agreeing the cut is the decision to stop looking and start sequencing, and it is what unblocks `mvp/BACKLOG.md` — one entry per feature with its dependencies, the data it needs, and the open questions that block it. Until then there is a specification and no build plan.

What is out, and why

Deferred	Reason
Planning and inventory (PL1–PL5, I1–I4)	Needs CMMS and stores integration depth that does not change whether the loop works
Alerts (L1–L6)	Valuable, but the MVP has few enough assets that alert noise is not yet the problem
Customer portal (R9, U1)	Multi-tenant surface area is a large security burden for little MVP learning
Executive briefing (C14, U2, R6)	Needs a history of outcomes to summarise; there will not be one yet
Watches (C13)	Depends on alerting
Report authoring and distribution (R2, R4, R7, R8)	R1 already answers a question and R5 already drills to source. Building reports, comparing scopes and scheduling delivery are reporting-product work, not proof that the loop closes
Work order logistics (W5–W7, W11, W12)	Scheduling, assignment, parts and SLA routing all need CMMS and stores depth. RC16 already puts a named person on the job deterministically, which is what the loop requires
Asset analytics (A2–A5)	Health scores, dependencies, like-for-like comparison and repeat-failure detection need a history of closed cases. There will not be one until the MVP has run
Knowledge beyond the SOP (K2, K3, K4, K6)	Manuals, work order history and verified cases are all corpora we do not yet have indexed. K1 and K5 prove retrieval with the source shown, which is the mechanism being demonstrated
Cooling tower (F13)	Blocked on S2, and the tower changes condenser water behaviour in ways that complicate early model validation
Re-baselining (F12)	Not needed until the first verified major repair completes
Break-glass, policy simulation (G7, G8)	Governance maturity, not product proof
Energy cost, forecasting, optimisation advice (E2–E4)	The analytics exist on the platform already; the MVP needs the baseline E1, not the advice layer
Business outcome check (V5)	Needs a measured energy or downtime improvement to appear, which takes longer than the MVP runs. V1 – V4 prove the repair worked; V5 proves it was worth doing
Planner queue and permission explanation (U4, U5)	Both are surfaces for personas the MVP does not include
Permit, qualification, EHS routing (S2, S3, S5)	All three are P0 and none is deferred on value — they need a permit system, a qualification record and an escalation matrix that are the customer's, not ours. S1 still blocks the unsafe action; what is deferred is the paperwork around it
Learning from closed cases (F9, V7)	Deliberately withheld: without a retraction mechanism one wrong confirmed root cause becomes permanent precedent. See <code>CONTEXT.md</code> §10
Multi-skill orchestration	C10 carries a job across steps without a planner running skills in parallel

Every one of the 147 registered features appears in exactly one of the two tables above. `scripts/verify.py` checks that against `mvp/FEATURE-REGISTER.md` and fails if a feature is in neither or in both.

The cases the MVP must handle

Fifteen. Each one is a question somebody actually asks, and the point of the list is coverage of *kinds* rather than volume — a happy path, a refusal, a contradiction, a safety stop, a data-quality lie, and a duplicate. The full flow and the feature IDs behind each are in `mvp/MVP.html` section 7; this is the source list.

#	Case	Kind
1	Manager finds today's important problems	happy path, read
2	Fault becomes a work order	the loop, forward
3	Technician gets help	field, scoped
4	Repair is verified	the loop, closing
5	Executive wants the business story	read, deferred surface
6	Somebody asks for something they may not have	authorization
7	The answer is "not enough evidence"	refusal
8	A detected fault nobody ever looked at	operational silence
9	The narrowing runs out of questions	exhausted, not settled
10	The machine has to stop now	safety, not a work order
11	The window was blind, and the report says so	data-quality honesty
12	Many alerts are really one problem	duplicate work orders — eight sub-cases
13	Two labels that contradict each other	inconsistent evidence
14	Same fault, two machines, only one model worth trusting	model trust
15	The machine is off, not broken	the cheapest false positive

Cases 12 to 15 were added from measurement rather than imagination. Each is visible in the measured window of our own database, and each shows the platform declining to be confident rather than performing well — which is the harder thing to demonstrate and the reason the honesty features are in the cut. See `docs/20-architecture/00-data-model.md` §4a–§4d.

MVP acceptance criteria

The MVP is done when all of these are demonstrably true on real site data:

1. A fault raised by the FDD engine can be explained by the Copilot with its residuals, gates and rule path visible.
2. When a gate fails, the Copilot returns `NO_DIAGNOSIS` and names the failed check — verified by deliberately invalidating a signal.

3. A work order created from that fault carries its evidence, and its priority can be recalculated by hand from the formula.
4. The work order cannot be closed until verification returns PASS.
5. Verification returns `UNKNOWN`, not `PASS`, when the machine has not run under comparable conditions since the work.
6. An out-of-scope request is blocked with a plain reason and the correct route.
7. Every material action has a complete audit trail with a policy version.
8. No answer in the acceptance run contains a number absent from the evidence pack.
9. A follow-up question — "and its ΔT ?" — resolves against the previous turn on **every** route, and every turn is present in the stored thread with no holes. Verified by asking the same follow-up after each kind of answer.
10. A case cannot leave `awaiting findings` while a blocking item is unanswered, and a "could not check" answer does **not** satisfy the item — verified by attempting both.
11. Escalating a case up leaves it unassigned and says so, rather than implying somebody has it.
12. Re-running the fault scan over the same window opens no second case for an episode that already has one.
13. Every checklist item shown to a technician traces to the curated library, and no item in the acceptance run was generated by a model.

Sequencing

Stage	Build	Depends on
0	Data in, tags normalised, asset hierarchy	—
1	Six models, residuals, validity flags, model health	Stage 0, Q1, Q2, Q14
2	Gates, persistence, isolation path, NO_DIAGNOSIS	Stage 1, Q3–Q6, Q9
3	Control Plane: identity, scope, risk, tool gateway, audit	Stage 0
4	Copilot read path: ask, context, route, evidence, explain	Stages 2 and 3
5	Work orders: create, evidence, priority, findings	Stages 3 and 4
6	Verification and the close gate	Stages 1, 2 and 5, plus Q15
7	Reports and knowledge retrieval	Stage 4
8	Conversation shell: threads, turn memory on every route, the conversational front door, read-only lookup	Stages 3 and 4
9	Case resolution: state machine, curated checklists, capability routing, the blocking gate, escalation routes	Stages 2, 5 and the checklist library being SME-reviewed

Stages 1 and 2 are the critical path and are the ones blocked by SME questions. Stages 3 and 8 have no SME dependency and can start immediately — the Control Plane and the conversation shell are both plain software.

Stage 9 has a dependency that is not a threshold: the checklist library is curated content and cannot ship until an SME has reviewed it, because a plausible-but-wrong checklist item directs physical work on pressurised equipment. That review is the long pole nobody costs for.

Four MVP features are named by no stage — A1 , U3 , S1 and S4 . Two of them are safety features. This is question Q17.